

Lecture 08 - Local Illumination

This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Source chunk	course_text_parts/03_lectures/08_local-illumination.txt
Extracted slide pages	62
Reader purpose	Commentary and causal explanation for human reading

How to read this lecture

- Read the section commentary before looking at the raw slide bullets.
- Use the mental model as the short version you should remember.
- Answer the check question without notes before continuing.

Per-slide commentary

Each extracted slide page gets its own note. Use this section as the spoken commentary that the terse slide deck is missing.

Page 1	Visual or title slide
Source cue	No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Professor-style explanation	The slide 'Visual or title slide' is primarily a visual, title, transition, or diagram page rather than a text-heavy concept slide. Its role is to place the next concept into the lecture flow and give a visual anchor for the topic that follows. On the slide, the concrete items are: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content. In the surrounding lecture sequence, this kind of page usually marks a shift of attention: from one pipeline stage to another, from a general idea to an algorithm, or from theory to an implementation detail.
Technical commentary	This slide is about Visual or title slide. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Visual or title slide' into a causal sentence instead of repeating the slide title?
Page 2	Local Illumination
Source cue	- Illumination is the process of simulating light interactions in a virtual scene / - Requires light and material properties / - Achieved through illumination model / - Essential for scene perception / without illumination with illumination

Professor-style explanation	The slide 'Local Illumination' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Illumination is the process of simulating light interactions in a virtual scene / - Requires light and material properties / - Achieved through illumination model / - Essential for scene perception / without illumination with illumination. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Local Illumination. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Illumination is the process of simulating light interactions in a virtual scene / - Requires light and material properties / - Achieved through illumination model / - Essential for scene perception / without illumination with illumination
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Local Illumination'?

Page 3	8.1 Physics of Light
Source cue	8.2 Light Sources / 8.3 Material Models / 8.4 Phong Illumination Model / 8.5 Shading
Professor-style explanation	The slide '8.1 Physics of Light' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: 8.2 Light Sources / 8.3 Material Models / 8.4 Phong Illumination Model / 8.5 Shading. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about 8.1 Physics of Light. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: 8.2 Light Sources / 8.3 Material Models / 8.4 Phong Illumination Model / 8.5 Shading
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.1 Physics of Light'?

Page 4	8.1 Physics of Light
Source cue	Understanding light in the real world
Professor-style explanation	The slide '8.1 Physics of Light' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: Understanding light in the real world. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about 8.1 Physics of Light. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: Understanding light in the real world
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.1 Physics of Light'?
Page 5	Electromagnetic Radiation
Source cue	- Electromagnetic radiation is defined by a spectrum of wavelengths / Wave length λ / - Visible light is subset of electromagnetic spectrum between 400 and 700nm, / which humans perceive as colors
Professor-style explanation	The slide 'Electromagnetic Radiation' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Electromagnetic radiation is defined by a spectrum of wavelengths / Wave length λ / - Visible light is subset of electromagnetic spectrum between 400 and 700nm, / which humans perceive as colors. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Electromagnetic Radiation. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Electromagnetic radiation is defined by a spectrum of wavelengths / Wave length λ / - Visible light is subset of electromagnetic spectrum between 400 and 700nm, / which humans perceive as colors
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Electromagnetic Radiation'?
Page 6	Visible Light
Source cue	1000 nm / Infrared / Radio / 700 nm / TV Red
Professor-style explanation	The slide 'Visible Light' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: 1000 nm / Infrared / Radio / 700 nm / TV Red. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Visible Light. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: 1000 nm / Infrared / Radio / 700 nm / TV Red
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Visible Light'?
Page 7	Visible Light
Source cue	1000 nm / Infrared / Radio / 700 nm / TV Red
Professor-style explanation	The slide 'Visible Light' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: 1000 nm / Infrared / Radio / 700 nm / TV Red. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Visible Light. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: 1000 nm / Infrared / Radio / 700 nm / TV Red
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Visible Light'?

Page 8	Light Spectra
Source cue	- Different light sources have different light spectra / - Intensity differs based on wavelength in visible spectrum / [http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html]
Professor-style explanation	The slide 'Light Spectra' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Different light sources have different light spectra / - Intensity differs based on wavelength in visible spectrum / [http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html]. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Spectra. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Different light sources have different light spectra / - Intensity differs based on wavelength in visible spectrum / [http://www.micro.magnet.fsu.edu/optics/lightandcolor/sources.html]
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Spectra'?
Page 9	Black-Body Radiation
Source cue	- A continuous spectrum emitted by a black body at a certain temperature / Steel / > 1570 K / 820 K / Stars
Professor-style explanation	The slide 'Black-Body Radiation' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: - A continuous spectrum emitted by a black body at a certain temperature / Steel / > 1570 K / 820 K / Stars. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Black-Body Radiation. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - A continuous spectrum emitted by a black body at a certain temperature / Steel / > 1570 K / 820 K / Stars
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.

Check yourself	Can you turn 'Black-Body Radiation' into a causal sentence instead of repeating the slide title?
Page 10	Black-Body Radiation
Source cue	Stars / https://www.tec-science.com/thermodynamics/temperature/black-body-radiation/ 23/06/2026 10
Professor-style explanation	The slide 'Black-Body Radiation' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: Stars / https://www.tec-science.com/thermodynamics/temperature/black-body-radiation/ 23/06/2026 10. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Black-Body Radiation. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: Stars / https://www.tec-science.com/thermodynamics/temperature/black-body-radiation/ 23/06/2026 10
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Black-Body Radiation' into a causal sentence instead of repeating the slide title?
Page 11	Spectral Density Function (SDF)
Source cue	- Light spectra are represented as spectral density functions / - Spectral density functions provide the light intensity (=energy) per wavelength / 400 Wavelength λ (nm) 700 / ygrenE / 400 Wavelength λ (nm) 700
Professor-style explanation	The slide 'Spectral Density Function (SDF)' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light spectra are represented as spectral density functions / - Spectral density functions provide the light intensity (=energy) per wavelength / 400 Wavelength λ (nm) 700 / ygrenE / 400 Wavelength λ (nm) 700. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Spectral Density Function (SDF). The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light spectra are represented as spectral density functions / - Spectral density functions provide the light intensity (=energy) per wavelength / 400 Wavelength λ (nm) 700 / ygrenE / 400 Wavelength λ (nm) 700

Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Spectral Density Function (SDF)'?
Page 12	Light Perception
Source cue	- Humans perceive light spectra / as differently colored lights / - Often described using three / main properties (HSL/HSV model) / - Hue - primary color tone
Professor-style explanation	The slide 'Light Perception' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Humans perceive light spectra / as differently colored lights / - Often described using three / main properties (HSL/HSV model) / - Hue - primary color tone. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Perception. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Humans perceive light spectra / as differently colored lights / - Often described using three / main properties (HSL/HSV model) / - Hue - primary color tone
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Perception'?
Page 13	Interpreting SDFs
Source cue	- Shape of an SDF depicts how humans perceive represented light / - Hue: dominant wavelength / - Saturation: % of energy in dom. wavelength / - Luminance: total energy (SDF integral) / 400 Wavelength λ (nm) 700

Professor-style explanation	The slide 'Interpreting SDFs' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Shape of an SDF depicts how humans perceive represented light - Hue: dominant wavelength - Saturation: % of energy in dom. wavelength - Luminance: total energy (SDF integral) / 400 Wavelength λ (nm) 700. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Interpreting SDFs. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Shape of an SDF depicts how humans perceive represented light - Hue: dominant wavelength - Saturation: % of energy in dom. wavelength - Luminance: total energy (SDF integral) / 400 Wavelength λ (nm) 700
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Interpreting SDFs'?
Page 14	Representing SDFs
Source cue	- In spectral rendering SDFs are discretized as arrays - Cell number represents differentiable wavelength ranges - Cell values represent energy per wavelength range - Much more frequently SDFs are represented as three-channel color - HSL values are derived from SDF
Professor-style explanation	The slide 'Representing SDFs' explains the problem of deciding which surface is actually visible from the current viewpoint. Many primitives can project to the same image region, but only the nearest relevant surface should determine the opaque color at a pixel. Different algorithms solve this relation in different spaces: object-space methods sort or split geometry, image-space methods subdivide regions, ray methods query visibility along viewing rays, and the depth buffer compares per-fragment depth values. On the slide, the concrete items are: - In spectral rendering SDFs are discretized as arrays - Cell number represents differentiable wavelength ranges - Cell values represent energy per wavelength range - Much more frequently SDFs are represented as three-channel color - HSL values are derived from SDF. The object-level relation is viewpoint, candidate surface, visibility rule, and visible result. This relation is the reason that fragment generation and final pixel color are not the same thing.

Technical commentary	This slide is about Representing SDFs. The visibility method on the slide decides which surface, fragment, ray hit, or image region is visible from the current viewpoint. Concrete items shown: - In spectral rendering SDFs are discretized as arrays / - Cell number represents differentiable wavelength ranges / - Cell values represent energy per wavelength range / - Much more frequently SDFs are represented as three-channel color / - HSL values are derived from SDF
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Representing SDFs' works per object, per image region, per ray, or per fragment?

Page 15	8.2 Light Sources
Source cue	How to represent light sources in computer graphics
Professor-style explanation	The slide '8.2 Light Sources' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: How to represent light sources in computer graphics. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about 8.2 Light Sources. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: How to represent light sources in computer graphics
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.2 Light Sources'?

Page 16	Light Sources
Source cue	- All light sources have a RGB color property representing their SDF / - Light sources are also defined through their location, orientation and extent / within a virtual scene / - Different light source types omit different spatial properties / - Light sources have no associated geometry

Professor-style explanation	The slide 'Light Sources' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - All light sources have a RGB color property representing their SDF / - Light sources are also defined through their location, orientation and extent / within a virtual scene / - Different light source types omit different spatial properties / - Light sources have no associated geometry. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Sources. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - All light sources have a RGB color property representing their SDF / - Light sources are also defined through their location, orientation and extent / within a virtual scene / - Different light source types omit different spatial properties / - Light sources have no associated geometry
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Sources'?

Page 17	Point Light Sources
Source cue	- Point light sources $\mathbf{L} = \mathbf{L}$, $\mathbf{L}$ have a geometric center $\mathbf{L} = (x, y, z)$ / and radiate light with color $\mathbf{L} = (\mathbf{L}, \mathbf{L}, \mathbf{L})$ evenly in all directions
Professor-style explanation	The slide 'Point Light Sources' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Point light sources $\mathbf{L} = \mathbf{L}$, $\mathbf{L}$ have a geometric center $\mathbf{L} = (x, y, z)$ / and radiate light with color $\mathbf{L} = (\mathbf{L}, \mathbf{L}, \mathbf{L})$ evenly in all directions. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Point Light Sources. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Point light sources $\mathbf{L} = \mathbf{L}$, $\mathbf{L}$ have a geometric center $\mathbf{L} = (x, y, z)$ / and radiate light with color $\mathbf{L} = (\mathbf{L}, \mathbf{L}, \mathbf{L})$ evenly in all directions
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Point Light Sources'?
Page 18	Cone Light Sources
Source cue	- Special point light source with geometric center / - Radiation of light restricted to a cone-shaped area with its tip in the center / - Specified by cut-off angle and exponents for the decrease / of brightness with distance to the center / (spot light exponent)
Professor-style explanation	The slide 'Cone Light Sources' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Special point light source with geometric center / - Radiation of light restricted to a cone-shaped area with its tip in the center / - Specified by cut-off angle and exponents for the decrease / of brightness with distance to the center / (spot light exponent). The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Cone Light Sources. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Special point light source with geometric center / - Radiation of light restricted to a cone-shaped area with its tip in the center / - Specified by cut-off angle and exponents for the decrease / of brightness with distance to the center / (spot light exponent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Cone Light Sources'?
Page 19	Light Attenuation
Source cue	- Light attenuation is inversely proportional to the square / of the distance ■ between the light source and the object / - Attenuation factor / ■ = min , 1 / ■■■ ■ + ■ · ■ + ■ · ■2

Professor-style explanation	The slide 'Light Attenuation' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light attenuation is inversely proportional to the square / of the distance d between the light source and the object / - Attenuation factor $f = \min(1, 1 / (k_a + k_d \cdot d + k_s \cdot d^2))$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Attenuation. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light attenuation is inversely proportional to the square / of the distance d between the light source and the object / - Attenuation factor $f = \min(1, 1 / (k_a + k_d \cdot d + k_s \cdot d^2))$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Attenuation'?

Page 20	Area Light Sources
Source cue	- Light source with geometric center and geometric extension / - Mostly of planar shape / - Transmission of light over the entire surface analogous to a point light source
Professor-style explanation	The slide 'Area Light Sources' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light source with geometric center and geometric extension / - Mostly of planar shape / - Transmission of light over the entire surface analogous to a point light source. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Area Light Sources. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light source with geometric center and geometric extension / - Mostly of planar shape / - Transmission of light over the entire surface analogous to a point light source

Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Area Light Sources'?
Page 21	Directed Light
Source cue	- Infinity-far away imaginary light source / - Light rays run parallel and evenly throughout the scene, / direction is specified as a vector / - Light intensity does not diminish / with increasing distance
Professor-style explanation	The slide 'Directed Light' explains the problem of deciding which surface is actually visible from the current viewpoint. Many primitives can project to the same image region, but only the nearest relevant surface should determine the opaque color at a pixel. Different algorithms solve this relation in different spaces: object-space methods sort or split geometry, image-space methods subdivide regions, ray methods query visibility along viewing rays, and the depth buffer compares per-fragment depth values. On the slide, the concrete items are: - Infinity-far away imaginary light source / - Light rays run parallel and evenly throughout the scene, / direction is specified as a vector / - Light intensity does not diminish / with increasing distance. The object-level relation is viewpoint, candidate surface, visibility rule, and visible result. This relation is the reason that fragment generation and final pixel color are not the same thing.
Technical commentary	This slide is about Directed Light. The visibility method on the slide decides which surface, fragment, ray hit, or image region is visible from the current viewpoint. Concrete items shown: - Infinity-far away imaginary light source / - Light rays run parallel and evenly throughout the scene, / direction is specified as a vector / - Light intensity does not diminish / with increasing distance
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.
Check yourself	Can you decide whether 'Directed Light' works per object, per image region, per ray, or per fragment?
Page 22	Local vs. Global Illumination
Source cue	- Simple case: Local lighting / - Consideration of direct illumination by light sources / (without obstacles or reflection) / - Simplified light models approximate / real lighting effect

Professor-style explanation	The slide 'Local vs. Global Illumination' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Simple case: Local lighting / - Consideration of direct illumination by light sources / (without obstacles or reflection) / - Simplified light models approximate / real lighting effect. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Local vs. Global Illumination. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Simple case: Local lighting / - Consideration of direct illumination by light sources / (without obstacles or reflection) / - Simplified light models approximate / real lighting effect
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Local vs. Global Illumination'?

Page 23	Ambient Light
Source cue	- Undirected light rays, evenly distributed / - Light intensity constant, regardless of distance / - Simulates general room brightness (basic brightness)
Professor-style explanation	The slide 'Ambient Light' explains the problem of deciding which surface is actually visible from the current viewpoint. Many primitives can project to the same image region, but only the nearest relevant surface should determine the opaque color at a pixel. Different algorithms solve this relation in different spaces: object-space methods sort or split geometry, image-space methods subdivide regions, ray methods query visibility along viewing rays, and the depth buffer compares per-fragment depth values. On the slide, the concrete items are: - Undirected light rays, evenly distributed / - Light intensity constant, regardless of distance / - Simulates general room brightness (basic brightness). The object-level relation is viewpoint, candidate surface, visibility rule, and visible result. This relation is the reason that fragment generation and final pixel color are not the same thing.
Technical commentary	This slide is about Ambient Light. The visibility method on the slide decides which surface, fragment, ray hit, or image region is visible from the current viewpoint. Concrete items shown: - Undirected light rays, evenly distributed / - Light intensity constant, regardless of distance / - Simulates general room brightness (basic brightness)
Why it matters	Visibility decides which generated candidates are actually seen from the current viewpoint.

Check yourself	Can you decide whether 'Ambient Light' works per object, per image region, per ray, or per fragment?
Page 24	8.3 Material Models
Source cue	How to represent materials in computer graphics
Professor-style explanation	The slide '8.3 Material Models' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: How to represent materials in computer graphics. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about 8.3 Material Models. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: How to represent materials in computer graphics
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.3 Material Models'?
Page 25	Material Models
Source cue	- Material models capture the different properties of materials / - Materials can have a wide variety of appearances / - Different groups of materials can be classified / based on appearance / - Metal – shiny and specular appearance
Professor-style explanation	The slide 'Material Models' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Material models capture the different properties of materials / - Materials can have a wide variety of appearances / - Different groups of materials can be classified / based on appearance / - Metal – shiny and specular appearance. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.

Technical commentary	This slide is about Material Models. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Material models capture the different properties of materials / - Materials can have a wide variety of appearances / - Different groups of materials can be classified / based on appearance / - Metal – shiny and specular appearance
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Material Models'?

Page 26	Materials vs. Textures
Source cue	- Material models specify the material properties for an infinitesimal surface / - No material variance with respect to location is supported / (i.e., structures such as wood grain cannot be represented) / - Textures are able to support material variance with respect to location
Professor-style explanation	The slide 'Materials vs. Textures' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Material models specify the material properties for an infinitesimal surface / - No material variance with respect to location is supported / (i.e., structures such as wood grain cannot be represented) / - Textures are able to support material variance with respect to location. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Materials vs. Textures. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Material models specify the material properties for an infinitesimal surface / - No material variance with respect to location is supported / (i.e., structures such as wood grain cannot be represented) / - Textures are able to support material variance with respect to location
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Materials vs. Textures'?

Page 27	Light Material Interaction 1/3
Source cue	- Material appearance is influenced by three wavelength-dependent effects / - Reflection – light energy bounces off / - Absorption – light energy is reduced / - Transmission – light passes through the object (refraction occurs) / R = reflected energy

Professor-style explanation	The slide 'Light Material Interaction 1/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Material appearance is influenced by three wavelength-dependent effects / - Reflection – light energy bounces off / - Absorption – light energy is reduced / - Transmission – light passes through the object (refraction occurs) / R = reflected energy. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Material Interaction 1/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Material appearance is influenced by three wavelength-dependent effects / - Reflection – light energy bounces off / - Absorption – light energy is reduced / - Transmission – light passes through the object (refraction occurs) / R = reflected energy
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 1/3'?
Page 28	Light Material Interaction 2/3
Source cue	- Material properties determine weight of the individual effects / opaque colored material opaque black material / $A > 0$, $T = 0$, $R > 0$ $A = I$, $T = 0$, $R = 0$ / perfect mirror material transparent material / $A = 0$, $T = 0$, $R = I$ $A \approx 0$, $R \approx 0$, $T \approx I$
Professor-style explanation	The slide 'Light Material Interaction 2/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Material properties determine weight of the individual effects / opaque colored material opaque black material / $A > 0$, $T = 0$, $R > 0$ $A = I$, $T = 0$, $R = 0$ / perfect mirror material transparent material / $A = 0$, $T = 0$, $R = I$ $A \approx 0$, $R \approx 0$, $T \approx I$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.

Technical commentary	This slide is about Light Material Interaction 2/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Material properties determine weight of the individual effects / opaque colored material opaque black material / $A > 0$, $T = 0$, $R > 0$ $A = 1$, $T = 0$, $R = 0$ / perfect mirror material transparent material / $A = 0$, $T = 0$, $R = 1$ $A \approx 0$, $R \approx 0$, $T \approx 1$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 2/3'?

Page 29	Light Material Interaction 3/3
Source cue	- Light-material interaction described by the Spectral Response Function (SRF) / 400 Wavelength λ (nm) 700 / ygrenE / 100% / All pass Filter (ideal, real)
Professor-style explanation	The slide 'Light Material Interaction 3/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light-material interaction described by the Spectral Response Function (SRF) / 400 Wavelength λ (nm) 700 / ygrenE / 100% / All pass Filter (ideal, real). The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Light Material Interaction 3/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light-material interaction described by the Spectral Response Function (SRF) / 400 Wavelength λ (nm) 700 / ygrenE / 100% / All pass Filter (ideal, real)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Light Material Interaction 3/3'?

Page 30	Reflection = SDF x SRF
Source cue	- Light-material interaction results are the product of the SDF and SRF / $100\% \times 100\%$ / SDF SRF / 0% 0% / 400 Wavelength λ (nm) 700 400 Wavelength λ (nm) 700

Professor-style explanation	The slide 'Reflection = SDF x SRF' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light-material interaction results are the product of the SDF and SRF / $100\% \times 100\%$ / SDF SRF / $0\% \ 0\%$ / 400 Wavelength λ (nm) 700 400 Wavelength λ (nm) 700. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Reflection = SDF x SRF. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light-material interaction results are the product of the SDF and SRF / $100\% \times 100\%$ / SDF SRF / $0\% \ 0\%$ / 400 Wavelength λ (nm) 700 400 Wavelength λ (nm) 700
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection = SDF x SRF'?

Page 31	Reflection Modeling 1/2
Source cue	- All light sources can be described by their Spectral Density Functions / - Natural light: sun, fire, ... / - Artificial light: light bulb, laser, neon tube, LED, ... / - All material properties can be described by Spectral Response Functions / - Glass, Water: SRF for transmission
Professor-style explanation	The slide 'Reflection Modeling 1/2' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - All light sources can be described by their Spectral Density Functions / - Natural light: sun, fire, ... / - Artificial light: light bulb, laser, neon tube, LED, ... / - All material properties can be described by Spectral Response Functions / - Glass, Water: SRF for transmission. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.

Technical commentary	This slide is about Reflection Modeling 1/2. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - All light sources can be described by their Spectral Density Functions / - Natural light: sun, fire, ... / - Artificial light: light bulb, laser, neon tube, LED, ... / - All material properties can be described by Spectral Response Functions / - Glass, Water: SRF for transmission
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection Modeling 1/2'?

Page 32	Reflection Modeling 2/2
Source cue	- Synthetic and natural objects are in general not self-emitting / - Light is reflected at its surfaces, i.e., re-emitted into the scene / - Reflection is dependent on surface properties / - Rough surface: scattering of light / - Smooth surface: reflection of light
Professor-style explanation	The slide 'Reflection Modeling 2/2' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Synthetic and natural objects are in general not self-emitting / - Light is reflected at its surfaces, i.e., re-emitted into the scene / - Reflection is dependent on surface properties / - Rough surface: scattering of light / - Smooth surface: reflection of light. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Reflection Modeling 2/2. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Synthetic and natural objects are in general not self-emitting / - Light is reflected at its surfaces, i.e., re-emitted into the scene / - Reflection is dependent on surface properties / - Rough surface: scattering of light / - Smooth surface: reflection of light
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Reflection Modeling 2/2'?

Page 33	8.4 Phong Illumination Model
Source cue	Determining the color of vertices

Professor-style explanation	The slide '8.4 Phong Illumination Model' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: Determining the color of vertices. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about 8.4 Phong Illumination Model. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: Determining the color of vertices
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to '8.4 Phong Illumination Model'?
Page 34	Empirical Observations
Source cue	- Light reflections are / - Dependent on the position of the observer / „... position of the observer...“ / - Dependent on the material properties of the surface / „... the specular properties of the object.“ Photo
Professor-style explanation	The slide 'Empirical Observations' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Light reflections are / - Dependent on the position of the observer / „... position of the observer...“ / - Dependent on the material properties of the surface / „... the specular properties of the object.“ Photo. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Empirical Observations. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Light reflections are / - Dependent on the position of the observer / „... position of the observer...“ / - Dependent on the material properties of the surface / „... the specular properties of the object.“ Photo
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Empirical Observations'?

Page 35	Illumination Models
Source cue	- Illumination models calculate the color of scene objects / based on the following properties / - Location – Position $\mathbf{p} = (x, y, z)$ of the selected point $\mathbf{p}$ / - Orientation – normal direction $\mathbf{n} = (n_x, n_y, n_z)$ of surface in $\mathbf{n} / x, y, z$
Professor-style explanation	The slide 'Illumination Models' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Illumination models calculate the color of scene objects / based on the following properties / - Location – Position $\mathbf{p} = (x, y, z)$ of the selected point $\mathbf{p}$ / - Orientation – normal direction $\mathbf{n} = (n_x, n_y, n_z)$ of surface in $\mathbf{n} / x, y, z$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Illumination Models. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Illumination models calculate the color of scene objects / based on the following properties / - Location – Position $\mathbf{p} = (x, y, z)$ of the selected point $\mathbf{p}$ / - Orientation – normal direction $\mathbf{n} = (n_x, n_y, n_z)$ of surface in $\mathbf{n} / x, y, z$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Illumination Models'?
Page 36	yaR
Source cue	thgiL / Diffuse Intensity 1/3 / - Simulation of light reflections on diffuse surfaces / - Model assumption / - Diffuse light reflections are non-directional (scattering)
Professor-style explanation	The slide 'yaR' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: thgiL / Diffuse Intensity 1/3 / - Simulation of light reflections on diffuse surfaces / - Model assumption / - Diffuse light reflections are non-directional (scattering). The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.

Technical commentary	This slide is about yaR. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: thgiL / Diffuse Intensity 1/3 / - Simulation of light reflections on diffuse surfaces / - Model assumption / - Diffuse light reflections are non-directional (scattering)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'yaR'?

Page 37	Diffuse Intensity 2/3
Source cue	- Equation / $I = k_d \cdot I_a \cdot \max(0, \cos \theta)$ / I_a diffuse material properties (surface dependent) / I_a diffuse light intensity (light source dependent)
Professor-style explanation	The slide 'Diffuse Intensity 2/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Equation / $I = k_d \cdot I_a \cdot \max(0, \cos \theta)$ / I_a diffuse material properties (surface dependent) / I_a diffuse light intensity (light source dependent). The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Diffuse Intensity 2/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Equation / $I = k_d \cdot I_a \cdot \max(0, \cos \theta)$ / I_a diffuse material properties (surface dependent) / I_a diffuse light intensity (light source dependent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Diffuse Intensity 2/3'?

Page 38	Diffuse Intensity 3/3
Source cue	$I = k_d \cdot I_a \cdot \max(0, \cos \theta)$ / I_a / https://bit.ly/2XFhbcM / Extrahierte Tabellen: / [Tabelle 1]

Professor-style explanation	The slide 'Diffuse Intensity 3/3' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: $I = I_a + I_r \cdot \cos^2(\theta)$ / $I_a = I_a \cdot \cos^2(\theta)$ / https://bit.ly/2XFhbcM / Extrahierte Tabellen: [Tabelle 1]. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Diffuse Intensity 3/3. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: $I = I_a + I_r \cdot \cos^2(\theta)$ / $I_a = I_a \cdot \cos^2(\theta)$ / https://bit.ly/2XFhbcM / Extrahierte Tabellen: [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Diffuse Intensity 3/3' into a causal sentence instead of repeating the slide title?

Page 39	Ambient Intensity 1/2
Source cue	- Simulation of the general room brightness / - Indirect light incidence or multiple reflections / - Model assumptions / - Constant influence on all scene objects / - No influence of position or alignment
Professor-style explanation	The slide 'Ambient Intensity 1/2' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Simulation of the general room brightness / - Indirect light incidence or multiple reflections / - Model assumptions / - Constant influence on all scene objects / - No influence of position or alignment. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Ambient Intensity 1/2. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Simulation of the general room brightness / - Indirect light incidence or multiple reflections / - Model assumptions / - Constant influence on all scene objects / - No influence of position or alignment
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Ambient Intensity 1/2'?

Page 40	Ambient Intensity 2/2
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Source cue	■ = ■ · ■ / ■■■■■■■■ ■ ■ / https://bit.ly/2XGn61n / Extrahierte Tabellen: / [Tabelle 1]
Professor-style explanation	The slide 'Ambient Intensity 2/2' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: ■ = ■ · ■ / ■■■■■■■■ ■ ■ / https://bit.ly/2XGn61n / Extrahierte Tabellen: / [Tabelle 1]. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Ambient Intensity 2/2. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: ■ = ■ · ■ / ■■■■■■■■ ■ ■ / https://bit.ly/2XGn61n / Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Ambient Intensity 2/2' into a causal sentence instead of repeating the slide title?
Page 41	Specular Intensity 1/3
Source cue	- Simulation of light reflections on specular surfaces / - Model assumption / - Reflection of light are directed (reflection) / - Observations / - Reflections of light are observer-dependent
Professor-style explanation	The slide 'Specular Intensity 1/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Simulation of light reflections on specular surfaces / - Model assumption / - Reflection of light are directed (reflection) / - Observations / - Reflections of light are observer-dependent. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Specular Intensity 1/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Simulation of light reflections on specular surfaces / - Model assumption / - Reflection of light are directed (reflection) / - Observations / - Reflections of light are observer-dependent
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Intensity 1/3'?

Page 42	Reflection Vector ■■
Source cue	- Calculation of the reflection vector / - Incidence angle corresponds to exit angle / - Equation / ■■ / ■■ = 2■ .. ■ · ■ – ■
Professor-style explanation	The slide 'Reflection Vector ■■' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: - Calculation of the reflection vector / - Incidence angle corresponds to exit angle / - Equation / ■■ / ■■ = 2■ .. ■ · ■ – ■. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Reflection Vector ■■. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: - Calculation of the reflection vector / - Incidence angle corresponds to exit angle / - Equation / ■■ / ■■ = 2■ .. ■ · ■ – ■
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Reflection Vector ■■' into a causal sentence instead of repeating the slide title?
Page 43	Specular Intensity 2/3
Source cue	- Equation / ■ = ■■■ · ■■■ · max(0,) / ■■■■■■■■ ■■ / - ■ specular material properties (surface dependant) / - ■ specular light intensity (light source dependent)
Professor-style explanation	The slide 'Specular Intensity 2/3' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Equation / ■ = ■■■ · ■■■ · max(0,) / ■■■■■■■■ ■■ / - ■ specular material properties (surface dependant) / - ■ specular light intensity (light source dependent). The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Specular Intensity 2/3. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Equation / ■ = ■■■ · ■■■ · max(0,) / ■■■■■■■■ ■■ / - ■ specular material properties (surface dependant) / - ■ specular light intensity (light source dependent)
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Intensity 2/3'?
Page 44	Specular Exponent ■
Source cue	- ■ influences fall off of specular highlight (■ ■ [1, ■] mit ■ ≈ 100) / ■ = 1 / ■■x(0, ■■ · ■■)1 / ■ = 5 / ■■x(0, ■■ · ■■)5
Professor-style explanation	The slide 'Specular Exponent ■' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - ■ influences fall off of specular highlight (■ ■ [1, ■] mit ■ ≈ 100) / ■ = 1 / ■■x(0, ■■ · ■■)1 / ■ = 5 / ■■x(0, ■■ · ■■)5. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Specular Exponent ■. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - ■ influences fall off of specular highlight (■ ■ [1, ■] mit ■ ≈ 100) / ■ = 1 / ■■x(0, ■■ · ■■)1 / ■ = 5 / ■■x(0, ■■ · ■■)5
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Specular Exponent ■'?
Page 45	Specular Intensity 3/3
Source cue	■■ = 2 · ■ · ■ · ■■ – ■■ / https://bit.ly/2XbPc7s ■ = ■ · ■ · ■■x(0, ■■ · ■■)■ / ■■■■■■■■■■ ■■ / Extrahierte Tabellen: / [Tabelle 1]
Professor-style explanation	The slide 'Specular Intensity 3/3' introduces a graphics object, operation, or relation that belongs to the rendering workflow. The named terms describe what data enters the step, what operation changes or classifies that data, and what result is passed onward. On the slide, the concrete items are: ■■ = 2 · ■ · ■ · ■■ – ■■ / https://bit.ly/2XbPc7s ■ = ■ · ■ · ■■x(0, ■■ · ■■)■ / ■■■■■■■■■■ ■■ / Extrahierte Tabellen: / [Tabelle 1]. The object-level relation is input, operation, output, and later use. This is the basic shape of most computer graphics concepts: data is represented in one form, processed by a rule or algorithm, and then consumed by the next stage of image generation.
Technical commentary	This slide is about Specular Intensity 3/3. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: ■■ = 2 · ■ · ■ · ■■ – ■■ / https://bit.ly/2XbPc7s ■ = ■ · ■ · ■■x(0, ■■ · ■■)■ / ■■■■■■■■■■ ■■ / Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.

Check yourself	Can you turn 'Specular Intensity 3/3' into a causal sentence instead of repeating the slide title?
Page 46	Blinn-Phong Illumination Model 1/2
Source cue	- Alternative simulation of specular reflections / - Replace mirror direction $\mathbf{r}$ with Halfway-Vector $\mathbf{h} / - \mathbf{h} = \mathbf{n} + \mathbf{r} / - \mathbf{h}$ is halfway between $\mathbf{n}$ and $\mathbf{r} / -$ Equation
Professor-style explanation	The slide 'Blinn-Phong Illumination Model 1/2' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Alternative simulation of specular reflections / - Replace mirror direction $\mathbf{r}$ with Halfway-Vector $\mathbf{h} / - \mathbf{h} = \mathbf{n} + \mathbf{r} / - \mathbf{h}$ is halfway between $\mathbf{n}$ and $\mathbf{r} / -$ Equation. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Blinn-Phong Illumination Model 1/2. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Alternative simulation of specular reflections / - Replace mirror direction $\mathbf{r}$ with Halfway-Vector $\mathbf{h} / - \mathbf{h} = \mathbf{n} + \mathbf{r} / - \mathbf{h}$ is halfway between $\mathbf{n}$ and $\mathbf{r} / -$ Equation
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Blinn-Phong Illumination Model 1/2'?
Page 47	Blinn-Phong Illumination Model 2/2
Source cue	$\mathbf{c} = \mathbf{a} + \mathbf{d} + \mathbf{r} / \text{https://bit.ly/2ICDwBD } \mathbf{r} = \mathbf{n} \cdot \mathbf{l} \cdot \mathbf{n} \times (0, \mathbf{n} \cdot \mathbf{l}) \mathbf{n} / \text{Extrahierte Tabellen: [Tabelle 1]}$
Professor-style explanation	The slide 'Blinn-Phong Illumination Model 2/2' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: $\mathbf{c} = \mathbf{a} + \mathbf{d} + \mathbf{r} / \text{https://bit.ly/2ICDwBD } \mathbf{r} = \mathbf{n} \cdot \mathbf{l} \cdot \mathbf{n} \times (0, \mathbf{n} \cdot \mathbf{l}) \mathbf{n} / \text{Extrahierte Tabellen: [Tabelle 1]}$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.

Technical commentary	This slide is about Blinn-Phong Illumination Model 2/2. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d + \mathbf{C}_s$ / https://bit.ly/2ICDwBD $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \mathbf{N} \cdot \mathbf{L} + \mathbf{C}_s \cdot \mathbf{R}^p$ / Extrahierte Tabellen: / [Tabelle 1]
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Blinn-Phong Illumination Model 2/2'?

Page 48	Phong Illumination Model
Source cue	- Additive combination of all factors / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d + \mathbf{C}_s$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \mathbf{N} \cdot \mathbf{L} + \mathbf{C}_s \cdot \mathbf{R}^p$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \max(0, \mathbf{N} \cdot \mathbf{L}) + \mathbf{C}_s \cdot \mathbf{R}^p$
Professor-style explanation	The slide 'Phong Illumination Model' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Additive combination of all factors / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d + \mathbf{C}_s$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \mathbf{N} \cdot \mathbf{L} + \mathbf{C}_s \cdot \mathbf{R}^p$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \max(0, \mathbf{N} \cdot \mathbf{L}) + \mathbf{C}_s \cdot \mathbf{R}^p$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Phong Illumination Model. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Additive combination of all factors / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d + \mathbf{C}_s$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \mathbf{N} \cdot \mathbf{L} + \mathbf{C}_s \cdot \mathbf{R}^p$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \max(0, \mathbf{N} \cdot \mathbf{L}) + \mathbf{C}_s \cdot \mathbf{R}^p$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Phong Illumination Model'?

Page 49	Phong Illumination Model
Source cue	- Additive combination of all factors / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d + \mathbf{C}_s$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \mathbf{N} \cdot \mathbf{L} + \mathbf{C}_s \cdot \mathbf{R}^p$ / $\mathbf{C} = \mathbf{C}_a + \mathbf{C}_d \cdot \max(0, \mathbf{N} \cdot \mathbf{L}) + \mathbf{C}_s \cdot \mathbf{R}^p$

Professor-style explanation	The slide 'Phong Illumination Model' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Additive combination of all factors $C = \frac{I_a}{\pi} + \frac{I_d}{\pi} \cdot \frac{1 + \cos \theta}{2} + \frac{I_s}{\pi} \cdot \frac{1 + \cos \theta_v}{2}$ = $I_a \cdot \frac{1}{\pi} + I_d \cdot \frac{1 + \cos \theta}{2} + I_s \cdot \frac{1 + \cos \theta_v}{2}$. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Phong Illumination Model. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Additive combination of all factors $C = \frac{I_a}{\pi} + \frac{I_d}{\pi} \cdot \frac{1 + \cos \theta}{2} + \frac{I_s}{\pi} \cdot \frac{1 + \cos \theta_v}{2}$ = $I_a \cdot \frac{1}{\pi} + I_d \cdot \frac{1 + \cos \theta}{2} + I_s \cdot \frac{1 + \cos \theta_v}{2}$
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Phong Illumination Model'?

Page 50	8.5 Shading
Source cue	Coloring triangles when using vertex-based illumination
Professor-style explanation	The slide '8.5 Shading' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: Coloring triangles when using vertex-based illumination. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about 8.5 Shading. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: Coloring triangles when using vertex-based illumination
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by '8.5 Shading'?
Page 51	Shading

Source cue	- Illumination model usually applied in the fragment stage / - Vertex-based illumination also possible to save compute / - Requires to spread the illumination information of the vertices / over the triangle / Flat Gouroud Phong
Professor-style explanation	The slide 'Shading' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Illumination model usually applied in the fragment stage / - Vertex-based illumination also possible to save compute / - Requires to spread the illumination information of the vertices / over the triangle / Flat Gouroud Phong. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Shading. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Illumination model usually applied in the fragment stage / - Vertex-based illumination also possible to save compute / - Requires to spread the illumination information of the vertices / over the triangle / Flat Gouroud Phong
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Shading'?
Page 52	Shading Computation
Source cue	- Tasks of shading procedures / - Colorize surfaces based on colors calculated by illumination model / - Shading is closely linked to the rasterization process and integrated into it / - Distribute lighting and shading / - Calculate light intensity for surface points
Professor-style explanation	The slide 'Shading Computation' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Tasks of shading procedures / - Colorize surfaces based on colors calculated by illumination model / - Shading is closely linked to the rasterization process and integrated into it / - Distribute lighting and shading / - Calculate light intensity for surface points. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.

Technical commentary	This slide is about Shading Computation. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Tasks of shading procedures / - Colorize surfaces based on colors calculated by illumination model / - Shading is closely linked to the rasterization process and integrated into it / - Distribute lighting and shading / - Calculate light intensity for surface points
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Shading Computation'?

Page 53	Flat Shading
Source cue	- Operation / - Calling the lighting model for a single point of a surface / (e.g., polygon center) / - Shading of the entire polygon in the determined intensity / - Useful if the following assumptions apply
Professor-style explanation	The slide 'Flat Shading' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Operation / - Calling the lighting model for a single point of a surface / (e.g., polygon center) / - Shading of the entire polygon in the determined intensity / - Useful if the following assumptions apply. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Flat Shading. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Operation / - Calling the lighting model for a single point of a surface / (e.g., polygon center) / - Shading of the entire polygon in the determined intensity / - Useful if the following assumptions apply
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Flat Shading'?

Page 54	Gouraud Shading 1/5
Source cue	- Shading method for polygons based on the interpolation / of intensity values at the polygon vertices / - Requirements / - Area decomposition in polygons (usually triangles) / - Normals for polygon corners (vertex normals)

Professor-style explanation	The slide 'Gouraud Shading 1/5' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Shading method for polygons based on the interpolation / of intensity values at the polygon vertices - Requirements - Area decomposition in polygons (usually triangles) - Normals for polygon corners (vertex normals). The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Gouraud Shading 1/5. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Shading method for polygons based on the interpolation / of intensity values at the polygon vertices - Requirements - Area decomposition in polygons (usually triangles) - Normals for polygon corners (vertex normals)
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 1/5'?

Page 55	Gouraud Shading 2/5
Source cue	- Proceeding / - Compute intensity values ■ for polygon corners ■ / ■ ■ / - Linear interpolation of the intensity values along the edges / - Linear interpolation of the intensity values along the scanlines
Professor-style explanation	The slide 'Gouraud Shading 2/5' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Proceeding / - Compute intensity values ■ for polygon corners ■ / ■ ■ / - Linear interpolation of the intensity values along the edges / - Linear interpolation of the intensity values along the scanlines. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Gouraud Shading 2/5. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Proceeding / - Compute intensity values ■ for polygon corners ■ / ■ ■ / - Linear interpolation of the intensity values along the edges / - Linear interpolation of the intensity values along the scanlines

Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 2/5'?
Page 56	Gouraud Shading 3/5
Source cue	- Shading at conceptually smooth transitions / - Geometrically "hard" edges instead of conceptually "soft" edges / - Gouraud shading can create visually smooth transitions / through smooth shading / - Example: ■ / ■ smooth transition, ■ / ■ hard transition
Professor-style explanation	The slide 'Gouraud Shading 3/5' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Shading at conceptually smooth transitions / - Geometrically "hard" edges instead of conceptually "soft" edges / - Gouraud shading can create visually smooth transitions / through smooth shading / - Example: ■ / ■ smooth transition, ■ / ■ hard transition. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Gouraud Shading 3/5. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Shading at conceptually smooth transitions / - Geometrically "hard" edges instead of conceptually "soft" edges / - Gouraud shading can create visually smooth transitions / through smooth shading / - Example: ■ / ■ smooth transition, ■ / ■ hard transition
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Gouraud Shading 3/5'?
Page 57	Gouraud Shading 4/5
Source cue	- interpolation problems / - The shading inside a triangle results from interpolation / - Intensity fluctuations in the interior are not taken into account / - Intensity fluctuations can only be considered by finer tessellation / - Example: Spotlight shines into the interior of a triangle

Professor-style explanation	The slide 'Gouraud Shading 4/5' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - interpolation problems / - The shading inside a triangle results from interpolation / - Intensity fluctuations in the interior are not taken into account / - Intensity fluctuations can only be considered by finer tessellation / - Example: Spotlight shines into the interior of a triangle. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Gouraud Shading 4/5. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - interpolation problems / - The shading inside a triangle results from interpolation / - Intensity fluctuations in the interior are not taken into account / - Intensity fluctuations can only be considered by finer tessellation / - Example: Spotlight shines into the interior of a triangle
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Gouraud Shading 4/5'?

Page 58	Gouraud Shading 5/5
Source cue	- Interpolation problems / - Contiguous polygons that do not fully share the same edges / will be shaded differently
Professor-style explanation	The slide 'Gouraud Shading 5/5' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Interpolation problems / - Contiguous polygons that do not fully share the same edges / will be shaded differently. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Gouraud Shading 5/5. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Interpolation problems / - Contiguous polygons that do not fully share the same edges / will be shaded differently
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.

Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Gouraud Shading 5/5'?
Page 59	Phong Shading 1/2
Source cue	- Interpolation of the corner normals instead of the corner intensities / - Proceeding / - Calculation of the corner normals / - Interpolation of the normals between the corners / - Interpolation between the endpoints of a scanline
Professor-style explanation	The slide 'Phong Shading 1/2' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Interpolation of the corner normals instead of the corner intensities / - Proceeding / - Calculation of the corner normals / - Interpolation of the normals between the corners / - Interpolation between the endpoints of a scanline. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Phong Shading 1/2. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Interpolation of the corner normals instead of the corner intensities / - Proceeding / - Calculation of the corner normals / - Interpolation of the normals between the corners / - Interpolation between the endpoints of a scanline
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Phong Shading 1/2'?
Page 60	Phong Shading 2/2
Source cue	- Advantages / - Pixel-based (i.e., image-precise) evaluation of the light sources / - Highly focused highlights at corners are not interpolated over the edge / - Highlights inside a triangle can be displayed / - Disadvantage

Professor-style explanation	The slide 'Phong Shading 2/2' explains the moment where continuous geometry becomes a discrete image problem. A mathematical line or triangle is not made of pixels, but the screen is a grid of samples. Rasterization decides which samples are covered by the projected primitive and creates fragments for those samples. At the same time, values attached to the primitive, such as depth, color, normals, or texture coordinates, are interpolated so that each fragment has the data needed for shading. On the slide, the concrete items are: - Advantages / - Pixel-based (i.e., image-precise) evaluation of the light sources / - Highly focused highlights at corners are not interpolated over the edge / - Highlights inside a triangle can be displayed / - Disadvantage. The important distinction is that rasterization creates candidates. A fragment exists because a primitive covered a sample, but final visibility still depends on depth testing, stencil testing, blending, masking, and framebuffer operations.
Technical commentary	This slide is about Phong Shading 2/2. The rasterization objects on the slide convert ideal geometric primitives into covered samples or fragments with interpolated attributes. Concrete items shown: - Advantages / - Pixel-based (i.e., image-precise) evaluation of the light sources / - Highly focused highlights at corners are not interpolated over the edge / - Highlights inside a triangle can be displayed / - Disadvantage
Why it matters	Rasterization determines fragment generation; without it, shading and fragment tests have nothing to operate on.
Check yourself	Can you explain which samples/fragments are generated by 'Phong Shading 2/2'?

Page 61	Literature and other sources used in this chapter
Source cue	No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Professor-style explanation	The slide 'Literature and other sources used in this chapter' is primarily a visual, title, transition, or diagram page rather than a text-heavy concept slide. Its role is to place the next concept into the lecture flow and give a visual anchor for the topic that follows. On the slide, the concrete items are: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content. In the surrounding lecture sequence, this kind of page usually marks a shift of attention: from one pipeline stage to another, from a general idea to an algorithm, or from theory to an implementation detail.
Technical commentary	This slide is about Literature and other sources used in this chapter. The slide names a concrete relation between input data, an operation, and an output that another graphics stage can consume. Concrete items shown: No object-level text was extracted from this page; the page is primarily title, image, diagram, or layout content.
Why it matters	Even a sparse slide usually names a relation you must be able to explain in words.
Check yourself	Can you turn 'Literature and other sources used in this chapter' into a causal sentence instead of repeating the slide title?

Page 62	Lesen und Ausprobieren
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Source cue	- Zum Lesen - Zum Ausprobieren / Marschner, Steve und Peter Shirley: / - Simulation von drei / Fundamentals of Computer Graphics, / ausgewählten Materialien
Professor-style explanation	The slide 'Lesen und Ausprobieren' explains how a visible surface point receives color from local lighting. The surface normal gives the orientation of the surface. The light vector gives the direction from which illumination arrives. The view vector connects the surface point to the observer, and the material parameters decide how strongly the surface reacts with ambient, diffuse, or specular reflection. On the slide, the concrete items are: - Zum Lesen - Zum Ausprobieren / Marschner, Steve und Peter Shirley: / - Simulation von drei / Fundamentals of Computer Graphics, / ausgewählten Materialien. The object-level relation is light source, surface point, normal, material response, and computed color. The visual behavior changes when any of these objects changes: a rotated normal changes diffuse brightness, a different view vector moves the specular highlight, and different material coefficients change the perceived surface type.
Technical commentary	This slide is about Lesen und Ausprobieren. The lighting objects on the slide combine normals, light directions, view directions, material coefficients, and shading locations to compute color. Concrete items shown: - Zum Lesen - Zum Ausprobieren / Marschner, Steve und Peter Shirley: / - Simulation von drei / Fundamentals of Computer Graphics, / ausgewählten Materialien
Why it matters	Lighting formulas are only meaningful when their vectors and material terms are interpreted correctly.
Check yourself	Can you identify the normal, light vector, view vector, and material term relevant to 'Lesen und Ausprobieren'?

08.1 Physics of Light

Source location: Section 8.1

Commentary

The physics slides provide intuition for light as energy interacting with surfaces. For this course, the goal is not full physical simulation but a usable model for real-time rendering.

Important terms are reflection, absorption, wavelength/color, intensity, and direction. These terms become shader inputs or material parameters later. The simplified model must still preserve the relation between light direction, surface orientation, and observed brightness.

Mental model

Shading is an approximation of light-surface interaction that is cheap enough for rendering.

Check yourself

Why does surface orientation affect diffuse brightness?

08.2 Light Sources

Source location: Section 8.2

Commentary

Light source models define where light comes from and how its direction and intensity are computed. Directional lights approximate distant sources with parallel rays. Point lights emit from a position and often include attenuation. Spotlights add a directional cone.

In shader code, the type of light determines how you compute the light vector. A directional light can use a fixed direction; a point light requires subtracting the surface position from the light position.

Mental model

Different light types mainly change how the light direction and intensity are computed at the surface point.

Check yourself

What is the difference between a directional light vector and a point light vector?

08.3 Material Models

Source location: Section 8.3

Commentary

A material model tells the shader how a surface responds to light. Diffuse material scatters light broadly; specular material creates view-dependent highlights; ambient terms approximate background illumination.

Material coefficients are not arbitrary decoration. They scale the contribution of each lighting term. If the specular coefficient is zero, the surface should not show a specular highlight under that model.

Mental model

Light asks what arrives; material answers how the surface reacts.

Check yourself

Which material parameter would you adjust to reduce shiny highlights?

08.4 Phong Illumination Model

Source location: Section 8.4

Commentary

The Phong illumination model combines ambient, diffuse, and specular terms. Diffuse lighting uses the angle between normal and light direction. Specular lighting also depends on the viewer because highlights move with the view direction.

The formula is less important than the decomposition. Ambient is a rough constant approximation. Diffuse is orientation-dependent matte reflection. Specular is view-dependent shininess. A correct answer should name the vectors and normalize them.

Mental model

Phong-style lighting is a sum of simple terms, each modeling a different visual effect.

Check yourself

Why does the specular term depend on the view direction?

08.5 Shading

Source location: Section 8.5

Commentary

Shading decides where the illumination calculation is evaluated and how results are interpolated. Flat shading uses one value per primitive. Gouraud shading evaluates at vertices and interpolates colors. Phong shading interpolates normals and evaluates lighting per fragment.

Per-fragment shading is more expensive but can represent highlights more accurately. This connects directly to the pipeline: interpolation during rasterization provides the data used by the fragment shader.

Mental model

The shading method decides what is computed at vertices and what is computed at fragments.

Check yourself

Why can Gouraud shading miss a small specular highlight?

End-of-lecture summary

If you remember only one thing from Lecture 08, remember this: This lecture explains how visible surface points receive color from local light, material, normal, and view information. It is about shading a point, not solving all global light transport.

Before moving on, you should be able to explain the lecture without reading slide bullets aloud. Use the vocabulary from the slides, but add causal links: why a step exists, what data it consumes, what it produces, and what can go wrong.